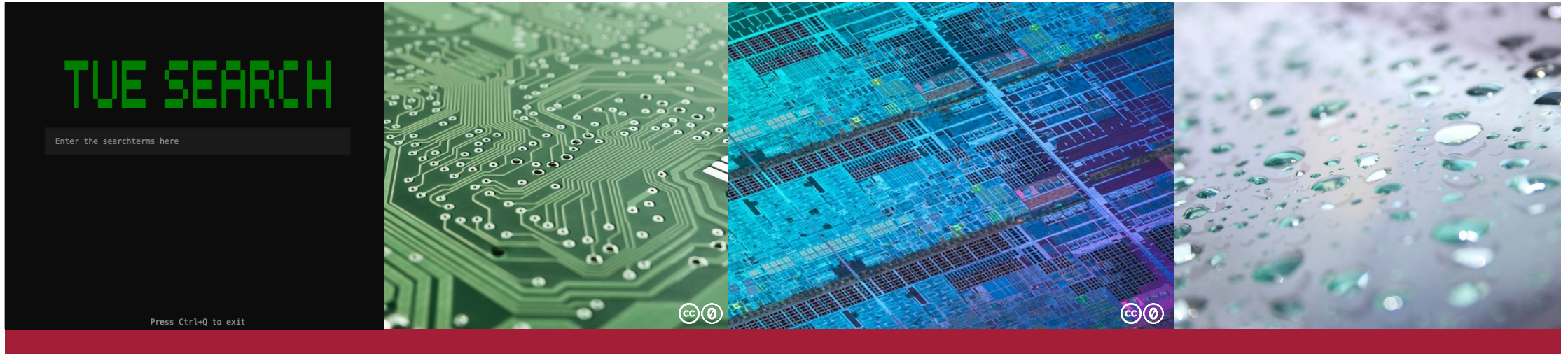


Modern Search Engines



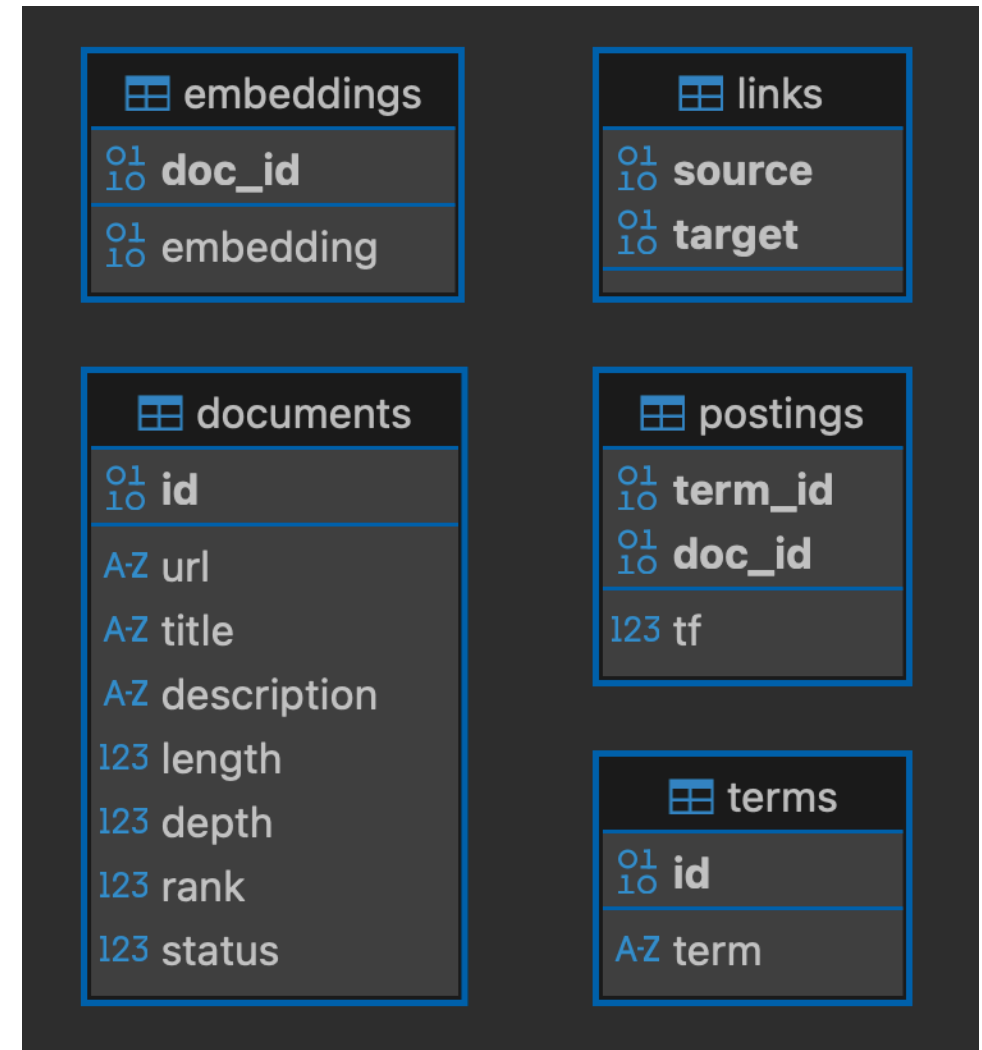
Search Engine Implementation

Group Project

06.08.2026 | Nico Henzler, Lukas Henzler, Tuba Aykan, Samuel Scheer

- Database
- Crawler and Indexer
- Ranking
- Live Demo and Batch Queries
- Summary

- SQLite as source of truth:
 - Document metadata
 - Crawl state
 - Terms and postings
- Storage and efficiency optimisations:
 - **Strict tables:**
Enforce types for consistency
 - **Deduplication:**
UNIQUE Key on normalised URLs
 - **SQL aggregation:**
Count matching docs within DB (TF-IDF)
 - **BLOBs:**
Store embedding float arrays as binary
 - **Only read what is required**



Crawler

- 12 parallel download workers
- Caches robots files
- Keeps status to resume after interrupts
 - Pending
 - Cached
 - Ready
 - Skipped
 - Error

PageRank

- Utilises stored link graph after crawling

Indexer

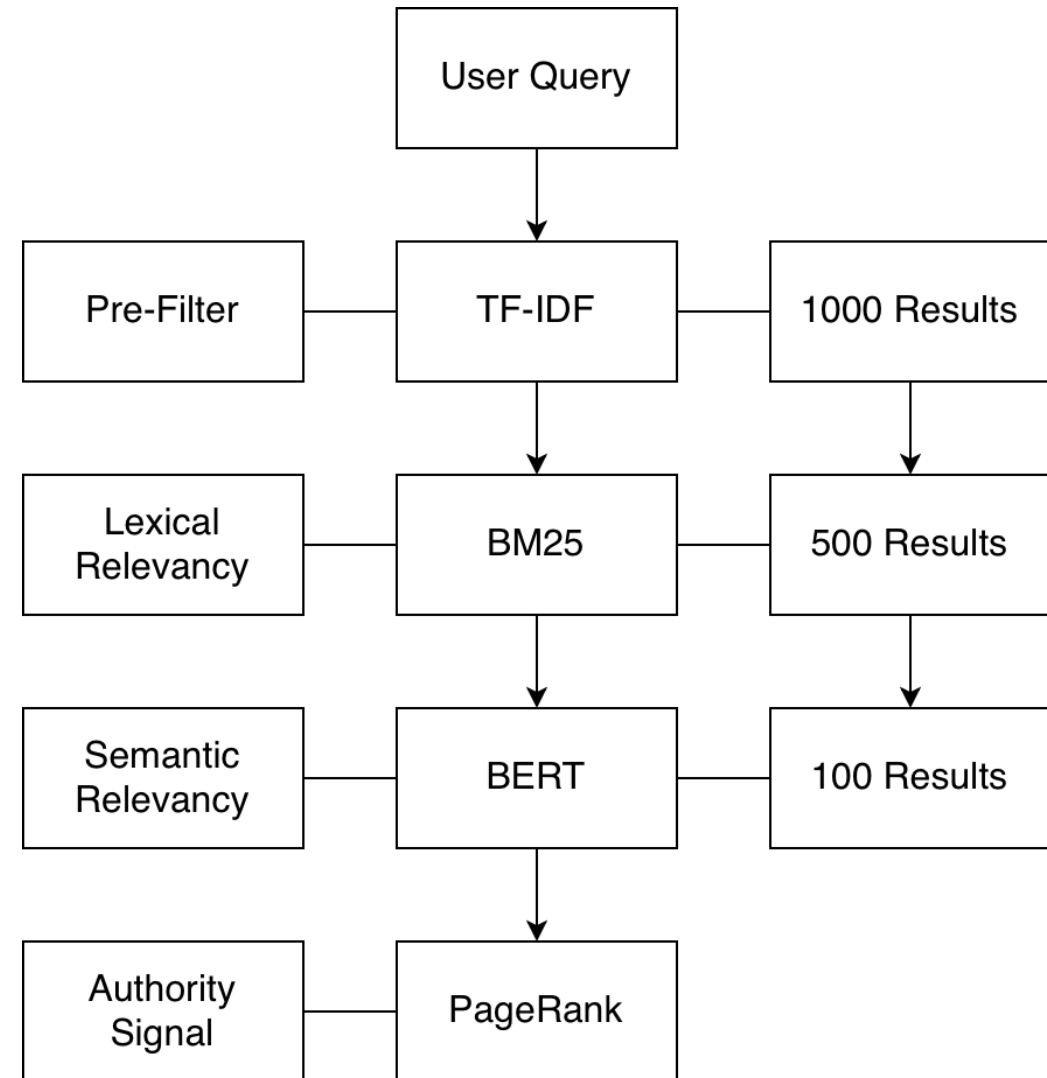
- 1 separate indexing worker
- Remove stop words
- Normalize umlauts
- Stemming was often too aggressive (using Porter2 and Snowball)
- Lemmatisation gave cleaner matching

BERT embeddings

- Once for each document based on:
 - Title, description, and content

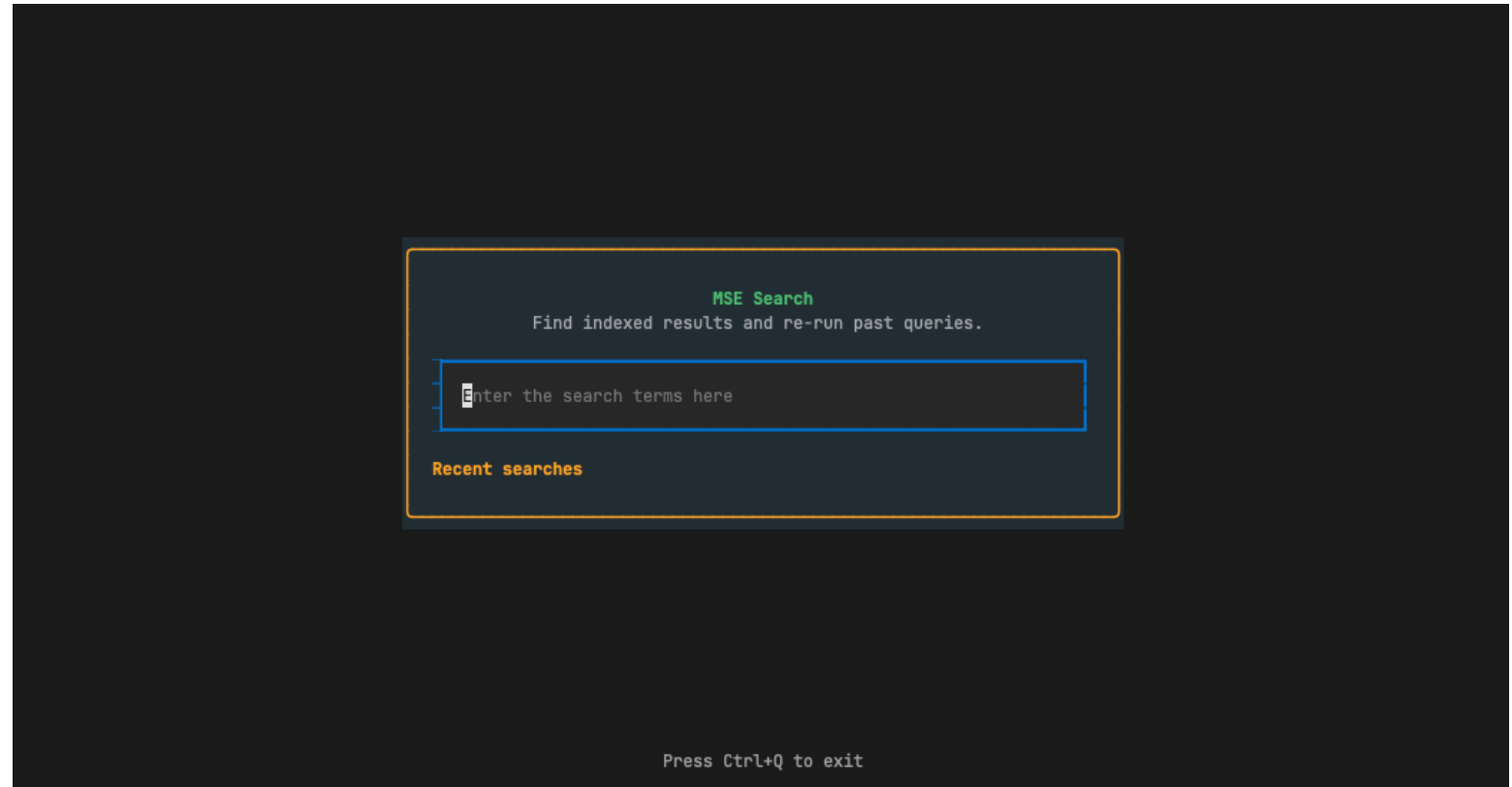
Fast-to-smart Funnel

- **Goal:**
Spend computation where it matters
- **Lightweight reduction:**
TF-IDF and BM25
- **Bi-encoder-style:**
Calculate BERT embeddings separately
- **PageRank:**
Boost well connected pages
- **Final score:**
 $0.7 \times \text{BERT} + 0.3 \times \text{PageRank}$



Batch Queries

LIVE DEMO



Summary

- Database as source of truth
- Persistent and restartable offline pipeline
- Preprocessing of PageRank and Document Embeddings
- Efficient lexical filtering at query time
- Semantic and authority-aware ranking

QUESTIONS?